

MATH310 Probability & Statistics

Section L2

Lecture 1: Axioms of Probability

Dr. Shafayat Abrar
Department of Electrical and Computer Engineering
Habib University, Karachi

Email: shafayat.abrar@sse.habib.edu.pk

<https://hulms.instructure.com/courses/3524>

Course Overview

- ▶ This course provides an introduction to probability and statistics.
- ▶ The first section will be on fundamentals of probability, including:
 - ▶ Sample space and Probability
 - ▶ Discrete and continuous random variables
 - ▶ Combinatorics (briefly)
 - ▶ Multiple random variables
 - ▶ Functions of random variables
 - ▶ Limit theorems (if time permits)
- ▶ The second section will be on statistics, including:
 - ▶ Parameter estimation
 - ▶ Hypothesis testing (if time permits)

Course materials

- ▶ Course syllabus, slides and homework assignments will be posted at <https://hulms.instructure.com/courses/3524>
- ▶ Grades will be posted at LMS and PSCS <https://pscs.habib.edu.pk/>
- ▶ The course text books are
 1. [Introduction to Probability](#), by Dimitri Bertsekas and John Tsitsiklis.
 2. [A First Course in Probability](#), by Sheldon Ross
 3. [Introduction to Probability](#), by Charles Grinstead and Laurie Snell

Assessment

- ▶ Two midterms (20% + 20%), and a final exam (25%).
- ▶ Activities and Homework (25%), Computer Assignment (10%)
- ▶ One small activity will be assigned after almost every lecture. Homework will be assigned upon completion one major topic.
- ▶ Homeworks should be submitted via Canvas by 5pm one week after it is assigned.

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- ▶ What does this mean?
- ▶ If I were to toss the coin 10 times, roughly 5 times I will see a head.
- ▶ A probability of 1 means it is certain, a probability of 0 means it is impossible.
- ▶ In general, you do an *experiment* many times, and you count how many times a particular *event* occurs. The proportion roughly gives you the probability of that particular event.

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- ▶ The different elements of a sample space must be **mutually exclusive** and **collectively exhaustive**.
 - ▶ Mutually exclusive means events are disjoint in nature, two or more events cannot occur at the same time.
 - ▶ Collectively exhaustive means if one of the events must occur.
- ▶ An event is a collection of possible outcomes.

Simple and compound events

- ▶ Simple event:
 - ▶ Your two coin tosses came up HH .
 - ▶ Your rolled die shows a 6
- ▶ Compound event: can be decomposed into simple events
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 - ▶ You got $(1, 1)$, or $(1, 3)$ or

Sets and sample spaces

We need to introduce some mathematical concepts to define probability more concretely:

- ▶ A **set** is a collection of objects, which are called **elements**
 - ▶ The natural numbers are a set, where the elements are individual numbers.
 - ▶ This class is the set, where the elements are the professor, the TA and the students.
- ▶ If an element x is in a set S , we write $x \in S$.
- ▶ If a set contains no elements, we call it the **empty set**, $\emptyset$.
- ▶ If a set contains every possible element, we call it the **universal set**, Ω .

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 - ▶ Set of positive integers = $\{1, 2, \dots\}$

Sets

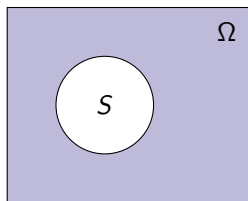
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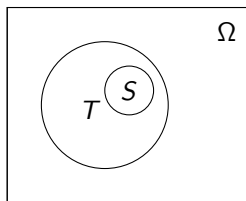
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 - ▶ the real numbers
 - ▶ the set of all subsets of natural numbers, aka the power set
- ▶ We can use curly brackets to describe a set in terms of its elements:
 - ▶ Sample space of a die roll: $S = \{1, 2, 3, 4, 5, 6\}$
 - ▶ Arbitrary set where all the elements meet some criterion C :
 $S = \{x | x \text{ satisfies } C\}$

Operations on sets

- ▶ Let the **universal set** Ω be the set of all objects we might possibly be interested in.
- ▶ The **complement**, S^c , of a set S , w.r.t. Ω , is the set of all elements that are in Ω but not in S . So $\Omega^c = \emptyset$.
- ▶ We say $S \subseteq T$, if every element in S is also in T .
- ▶ $S \subseteq T$ and $T \subseteq S$ if and only if $S = T$.



S^c is the shaded region



$S \subset T \subset \Omega$

Operations on sets: Union, Intersection, Difference

- ▶ The **union**, $S \cup T$, of two sets S and T is the set of elements that are in either S or T (or both): $S \cup T = \{x | x \in S \text{ or } x \in T\}$.

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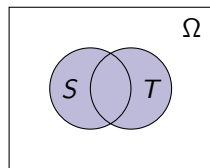
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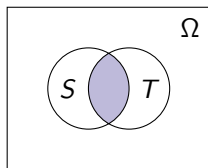
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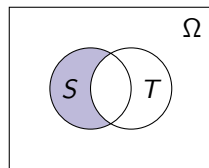
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$$S \cup T$$



$$S \cap T$$



$$S \setminus T = S \cap T^c$$

Operations on sets

- ▶ We can extend the notions of union and intersection to multiple (even infinitely many!) sets:

$$\bigcup_{i=1}^n S_i = S_1 \cup S_2 \cup \cdots \cup S_n = \{x \mid x \in S_i \text{ for some } 1 \leq i \leq n\}$$

$$\bigcap_{i=1}^n S_i = S_1 \cap S_2 \cap \cdots \cap S_n = \{x \mid x \in S_i \text{ for all } 1 \leq i \leq n\}$$

- ▶ We say two sets are **disjoint** if their intersection is empty.
- ▶ We say a collection of sets are disjoint if no two sets have any common elements.
- ▶ If a collection of disjoint sets have union S , we call them a **partition** of S .

Probability laws

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- ▶ Axioms of probability:
 - ▶ **Nonnegativity:** $P(A) \geq 0$, for every event A .
 - ▶ **Additivity:** If A and B are two disjoint events, then the probability of their union satisfies $P(A \cup B) = P(A) + P(B)$.
This extends to the union of infinitely many disjoint events:

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

- ▶ **Normalization:** The probability of the entire sample space Ω is equal to 1, i.e. $P(\Omega) = 1$

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This is an example of an **uniform distribution**, where all outcomes are equally likely.

Properties of probability laws

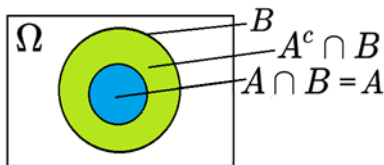
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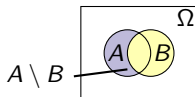
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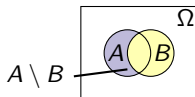


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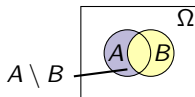
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